

ITA0609

```
# Import required libraries
```

```
from sklearn.datasets import load_iris
```

```
from sklearn.mixture import GaussianMixture
```

```
# Load the Iris dataset
```

```
iris = load_iris()
```

```
X = iris.data
```

Create the Gaussian Mixture Model (EM Algorithm)

```
gmm = GaussianMixture(n_components=3, random_state=42)
```

```
# Train the model
```

gmm.fit(X)

```
# Predict cluster labels
```

```
labels = gmm.predict(X)
```

```
# Display results
```

```
print("Cluster Labels:")
```

```
print(labels)
```

```
print("\nMeans of Clusters:")
```

```
print(gmm.means_)
```

Output:

```
... Cluster Labels:  
[1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1  
 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 2 2 2 2 2 2 2 2 2 0 2 0 2 0 2  
 2 2 2 0 2 2 2 2 2 0 2 2 2 2 2 2 2 2 2 2 2 2 2 0 0 0 0 0 0 0 0  
 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0  
 0 0]  
  
Means of Clusters:  
[[6.54639415 2.94946365 5.48364578 1.98726565]  
 [5.006      3.428      1.462      0.246      ]  
 [5.9170732  2.77804839 4.20540364 1.29848217]]
```